Seat No : ____________

[72/A-18]

No. of Printed Pages : 04

SARDAR PATEL UNIVERSITY

FYBCA – 2nd Semester (Under CBCS) 2020

US02FBCA02 | | Mathematics–II

Date: 10/02/2020 **Marks : 70**

Time : 2.00 pm to 4.00 pm. **No. of Printed Pages : 4**

Q-1 Select the correct option for each of the followings: **10**

1 Edges connecting the same endpoints are called ______________.

(a) multiple edges (b) multigraphs (c) loops (d) trivial graphs

2 A graph with one vertex and no edges is called ______________.

(a) empty graph (b) trivial graph (c) null graph (d) none of these

3 The degree of isolated vertex is ___________.

(a) one (b) null (c) zero (d) two

4 The chromatic number of a graph K_{120} is ______________.

(a) 40 (b) 30 (c) 100 (d) 120

5 In a connected map has V = 5 and R = 3 then E = ___________.

(a) 6 (b) 2 (c) 4 (d) 8

6 We can select 4 objects from the given 9 objects in ___________ ways.

(a) $\binom{9}{13}$ (b) $\binom{9}{4}$ (c) $\frac{9!}{4!}$ (d) none of these

7 $0! + 1! =$ ________________.

(a) 1 (b) 0 (c) 2 (d) none of these

8 $P(6, -3) =$ ___________.

(a) 120 (b) –18 (c) –120 (d) none

9 The quartile Q_2 coincides with ______________.

(a) Median (b) Mean (c) Mode (d) Standard Deviation

10 _________is the measure of dispersion.

(a) Mean (b) Standard Deviation (c) Mode (d) Median

Q-2 Do as directed (Attempt Any 10): **20**

1 Define the terms: null graph and simple path.

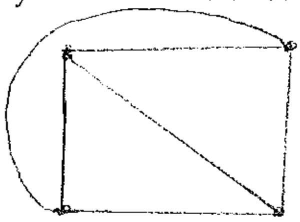
2 Draw diagram for the graphs:

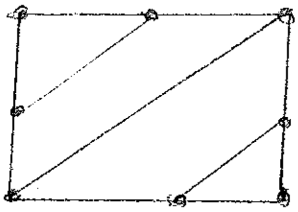
(a) V = {A, B, C, D} and E = [{A, B}, {D, A}, {C, A}, {C, D}]
(b) V = {P1, P2, P3, P4} and E = [{P1, P2}, {P2, P2}, {P2, P3}, {P2, P4}]

(P.T.O.)

3 Draw the diagram for the graphs K_4 and K_5.

4 Verify Euler's formula for the following:





5 Define chromatic number of a graph and find the chromatic number for the graph K_{10}.

6 Define planar graph and coloring a graph.

7 Simplify $\dfrac{(n-1)!}{(n+2)!}$

8 Find n if $P(n, 2) = 72$

9 A farmer buys 3 cows, 2 pigs and 2 hens from a man who has 6 cows, 5 pigs and 8 hens. How many choices does the farmer have?

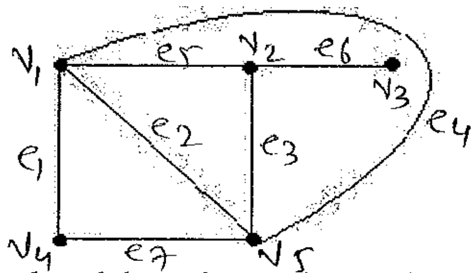
10 Write down the formula for Q_1 and Q_3.

11 Find the variance for the observations : 7, 7, 7, 7, 7, 7, 7.

12 Find the standard deviation for the data: 1, 2, 3, 4, 5.

Q-3 Do as directed.

A Find the adjacency and incidence matrix for the following graph. 6



B Define regular graph and draw the regular graph of degrees 1, 2 and 3. 4

OR

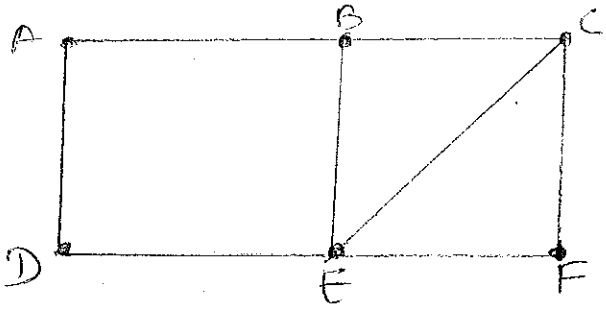
Q-3 Do as directed.

A Draw the graph G corresponding to each adjacency matrix given below 6

$$A = \begin{array}{ccccc} 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \\ 1 & 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 1 & 0 \end{array} \qquad A = \begin{array}{cccc} 1 & 1 & 2 & 0 \\ 1 & 2 & 1 & 3 \\ 2 & 1 & 0 & 1 \\ 0 & 3 & 1 & 0 \end{array}$$

B Consider the following graph G and find the followings: **4**

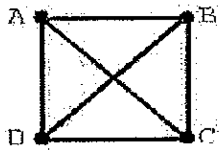
(a) G – A (b) G – B (c) G – C (d) G – D



Q-4 Do as directed.

A Write down Welch Powell algorithm to paint a graph and give one example of it. **5**

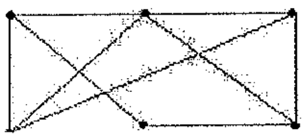
B Draw 5 spanning trees for the following graph. **5**

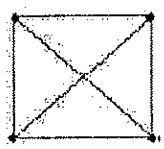


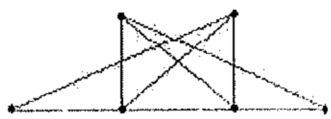
OR

Q-4 Do as directed.

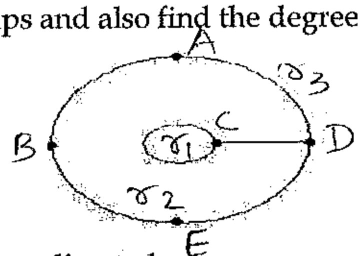
A Write down Euler's formula and draw the planar graph for the followings: **4**

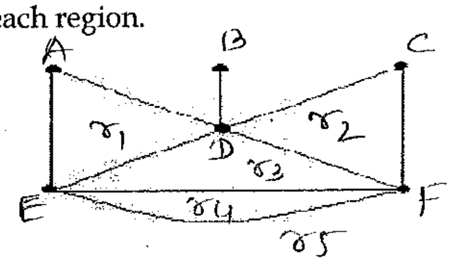






B Identify cycle or closed path that borders each region of the following maps and also find the degree of each region. **6**





Q-5 Do as directed.

A Suppose repetitions are not permitted. Find the followings also justify your answer. **6**

(a) The number of three digits numbers that can be formed from the six digits 2, 3, 5, 6, 7 and 9.

(b) How many of them are less than 400?

(c) How many are multiple of 5?

(P.T.O.)

B Find n if (a) 2P (n, 2) + 50 = P(2n, 2) (b) P(n, 4) = 42 P(n, 2) 4

OR

Q-5 Do as directed.

A Prove that $\binom{17}{6} = \binom{16}{5} + \binom{16}{6}$ 5

B The English alphabet has 21 consonants and 5 vowels. 5

(1) Find the number m of 5 letter words which contain 3 different consonants and 2 different vowels.
(2) Solve problem 1 if the words must contain the letter B.
(3) Solve problem 1 if the words must contain the letters B and C.
(4) Solve problem 1 if the words must begin with B and end with C.
(5) Solve problem 1 if the words must contain the letters A and B.

Q-6 From the following data find: 10

(1) Two regression coefficients.
(2) Two regression equations
(3) Most likely marks in Statistics when marks in Economics are 30

Marks in Economics	25	28	35	32	31	36	29	38	34	32
Marks in Statistics	43	46	49	41	36	32	31	30	33	39

OR

Q-6 Calculate Bowley's coefficient of skewness for the following data. 10

Weight (lbs)	No. of persons
0-99	1
100-109	14
110-119	66
120-129	122
130-139	145
140-149	121
150-159	65
160-169	31
170-179	12
180-189	5
190-199	2
200-209	2

—X—